

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 3: Airspeed Indicators & The Machmeter

Syllabus Source: CAE Oxford (pp. 79-118) | Pass Target: >= 90% AviationExam

1. AIRSPEED INDICATOR (ASI) PRINCIPLES & AIRSPEED HIERARCHY

The Airspeed Indicator measures dynamic pressure ($q = \frac{1}{2} \rho V^2$). Pitot pressure feeds the inside of an expandable metal capsule, while static pressure feeds the sealed instrument case: Differential Pressure = $P_{\text{pitot}} - P_{\text{static}} = \text{Dynamic Pressure } (q)$:

- Indicated Airspeed (IAS): The raw airspeed reading displayed on the cockpit ASI dial or PFD speed tape, uncorrected for errors.
- Calibrated Airspeed (CAS / RAS): IAS corrected for instrument errors and position / pressure errors (static vent aerodynamic disturbance). CAS is equal to EAS at low sub-transonic airspeeds at sea level.
- Equivalent Airspeed (EAS): CAS corrected for atmospheric compressibility effects. Crucial at high speeds and altitudes (> 250 kt / > 10,000 ft) where compressed air molecules generate false excess dynamic pressure: $EAS = CAS - \text{Compressibility Correction}$.
- True Airspeed (TAS): The actual physical velocity of the aircraft relative to the surrounding air mass. Calculated by correcting EAS for actual ambient air density: $TAS = EAS \times \sqrt{\rho_0 / \rho}$. TAS increases with altitude for a constant IAS (~1.5% to 2% increase per 1,000 ft)!

2. ASI COLOR CODING ARCS (CS-23 / CS-25)

Color Arc / Line	Speed Range Limits	Aerodynamic Significance & Pilot Actions
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White Arc	V_{SO} to V_{FE}	Flap operating range. Lower limit is stall speed in landing configuration (V_{SO}); upper limit is maximum flap extended speed (V_{FE}).
Green Arc	V_{S1} to V_{NO}	Normal operating range. Lower limit is stall speed clean (V_{S1}); upper limit is Maximum Structural Cruising Speed (V_{NO}).
Yellow Arc	V_{NO} to V_{NE}	Caution range. Maneuvers allowed in smooth air only; do not make abrupt control movements.
Red Radial Line	V_{NE} (Never Exceed)	Never Exceed speed. Exceeding risks catastrophic structural flutter and aerodynamic breakup!

3. THE MACHMETER: MECHANISM & OPERATIONAL ERRORS

Mach number is the ratio of True Airspeed (TAS) to the local speed of sound (a): $M = TAS / a$.
Local speed of sound depends EXCLUSIVELY on absolute temperature: $a = 38.94 \times \sqrt{T \text{ [Kelvin]}}$:

- Machmeter Architecture: Contains TWO separate capsules mechanically coupled via a floating sliding ratio divider arm: 1. Airspeed capsule (fed by pitot/static differential, sensing q); 2. Altimeter aneroid capsule (evacuated, sensing static ambient pressure P_s).
- Aeronautical Ratio: Because dynamic pressure q is proportional to $P_s \times M^2$, the ratio of $(P_t - P_s) / P_s$ depends ONLY ON MACH NUMBER! The altimeter capsule continuously modifies the pivot leverage of the airspeed capsule.

- Zero Temperature / Density Error: CRITICAL EXAM POINT: The Machmeter suffers NO DENSITY ERRORS and NO TEMPERATURE ERRORS! Temperature and density factors mathematically cancel out in the ratio of dynamic to static pressure. The Machmeter is subject ONLY to instrument error and position error!

[!] EXAM TRAP (AVIATIONEXAM) - Climbing / Descending at Constant Speed Scenarios

- CLIMB at Constant IAS: TAS INCREASES (density drops); Mach number INCREASES (temperature drops)! - CLIMB at Constant Mach: TAS DECREASES (until tropopause, then constant); IAS DECREASES rapidly! - CROSSOVER ALTITUDE: The altitude where the aircraft transitions from climbing at constant IAS to climbing at constant Mach number (typically FL 280 to FL 310).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Climb Rules for IAS vs Mach

Rule 1 (Climb at Constant IAS): Air density drops -> TAS increases to maintain dynamic pressure. Temperature drops -> speed of sound 'a' drops. Increasing TAS divided by decreasing 'a' means Mach number INCREASES. Rule 2 (Climb at Constant Mach): Above crossover altitude, climbing at constant Mach means dynamic pressure q must decrease as static pressure drops. Therefore IAS DECREASES. Summary: In climb: Constant IAS -> Mach climbs. Constant Mach -> IAS decays.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What errors affect a mechanical aircraft Machmeter?

- [A]: Density error and temperature error
- [B]: Instrument error and position (pressure) error ONLY
- [C]: Compressibility error only
- [D]: Lag error only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Machmeter measures the ratio of dynamic pressure to static pressure. Air density and temperature variables mathematically cancel out. Therefore, a Machmeter has NO density or temperature errors!

Q2: As an aircraft climbs at a constant Indicated Airspeed (IAS) of 280 knots from sea level up to FL 280, what

- [A]: Both TAS and Mach number decrease
- [B]: Both TAS and Mach number INCREASE
- [C]: TAS increases while Mach number remains constant
- [D]: TAS remains constant while Mach number increases

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. As altitude increases, air density drops, so TAS must increase for the same dynamic pressure (IAS). Simultaneously, outside air temperature drops, lowering the local speed of sound; therefore, Mach number increases rapidly!

Q3: What is the aerodynamic definition of Equivalent Airspeed (EAS)?

- [A]: IAS corrected for density altitude

- [B]: Calibrated Airspeed (CAS) corrected for compressibility effects
- [C]: Groundspeed corrected for wind
- [D]: True airspeed at the tropopause

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Equivalent Airspeed represents Calibrated Airspeed corrected for the adiabatic compressibility of air molecules at high forward speeds and high altitudes.